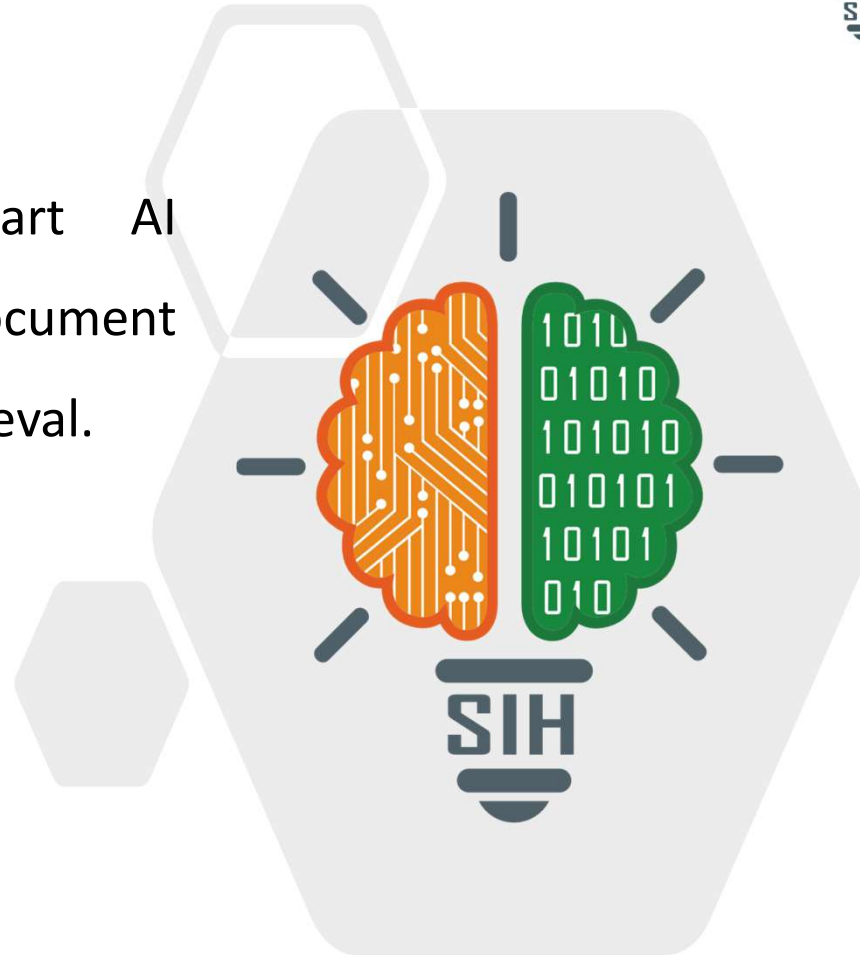


SMART INDIA HACKATHON 2026



- **Problem Statement Title-** Smart AI Platform for Multilingual Legal Document Classification, Summarization and Retrieval.
- **Theme –** Smart AI Platform
- **PS Category-** Software
- **Team Name -** LexAI



India's Legal System – The Problem

A growing backlog. Complex documents. A need for intelligent solutions.



SMART INDIA
HACKATHON
2026



Over 5 Crore Pending Cases

India's legal system faces a significant backlog

Major Contributor



Manual Effort

Processing complex and voluminous legal documents requires significant manual effort.



Complex & Voluminous Legal Documents

Legal documents are lengthy and difficult to analyze.



Multiple Indian Languages

Documents exist in various regional languages, making processing even more challenging.



No Scalable AI Platform

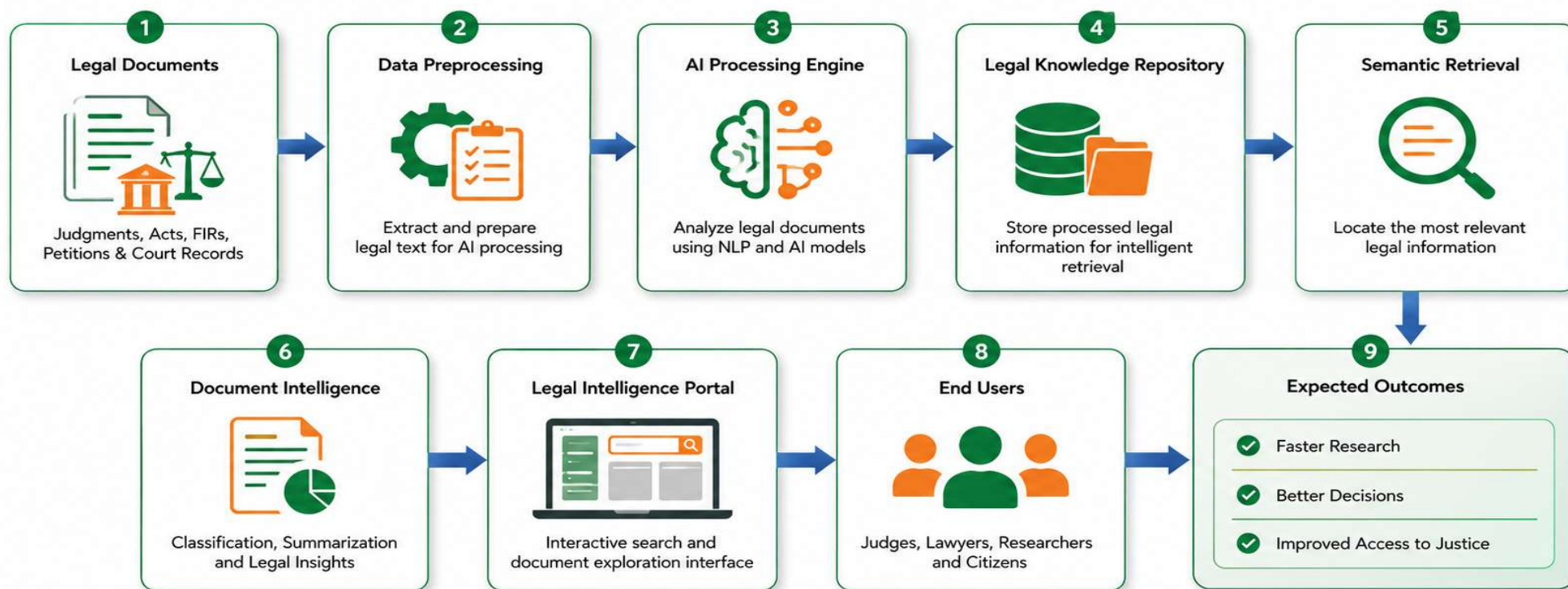
Currently, no AI platform exists that can classify and summarize legal documents in a multilingual Indian context.



The Result: Delays in justice delivery, inefficient workflows, and difficulty in accessing critical legal information.

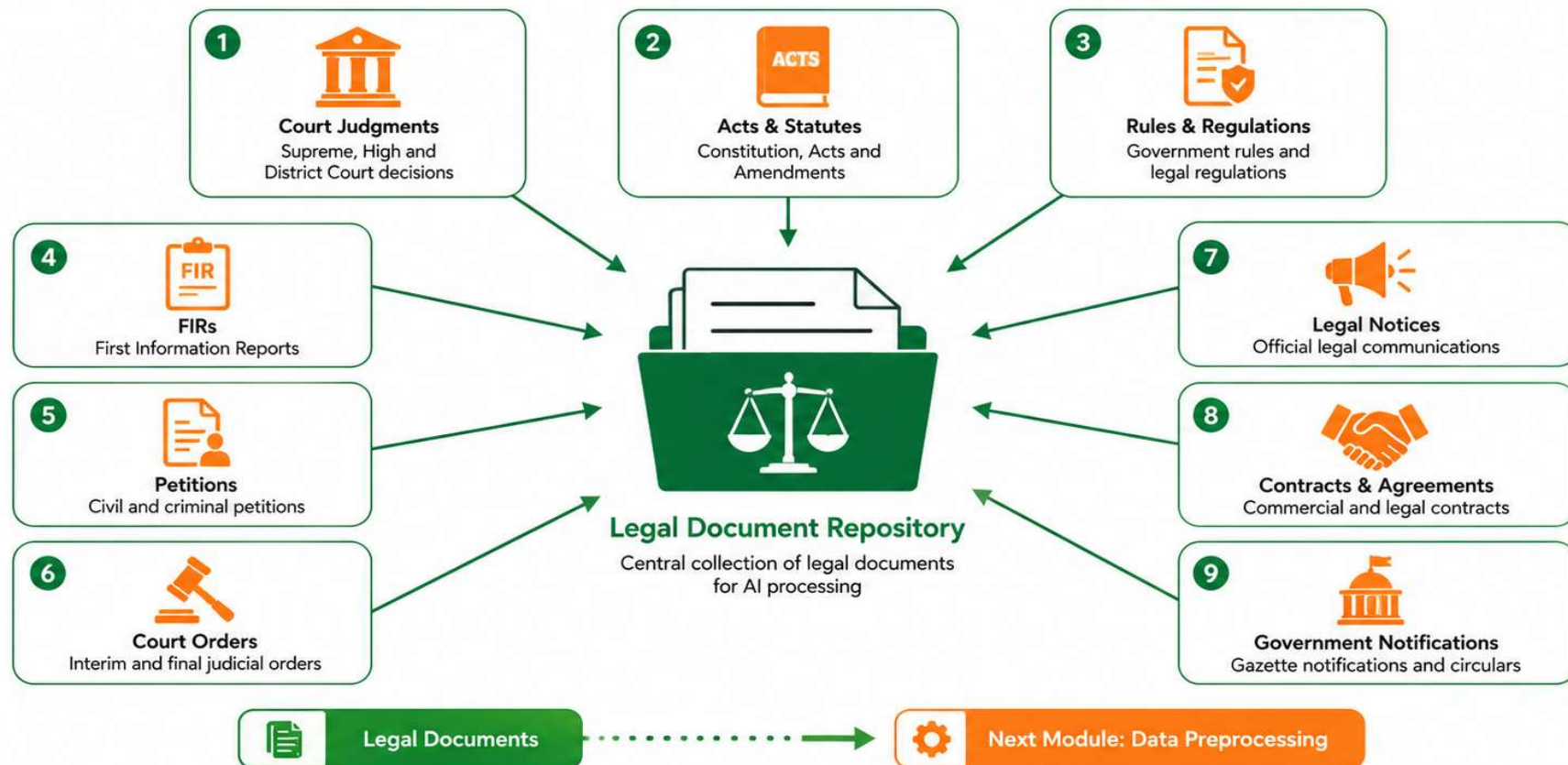
Proposed AI Powered Legal Document Intelligence System

for Automated Classification and Summarization in Indian Courts



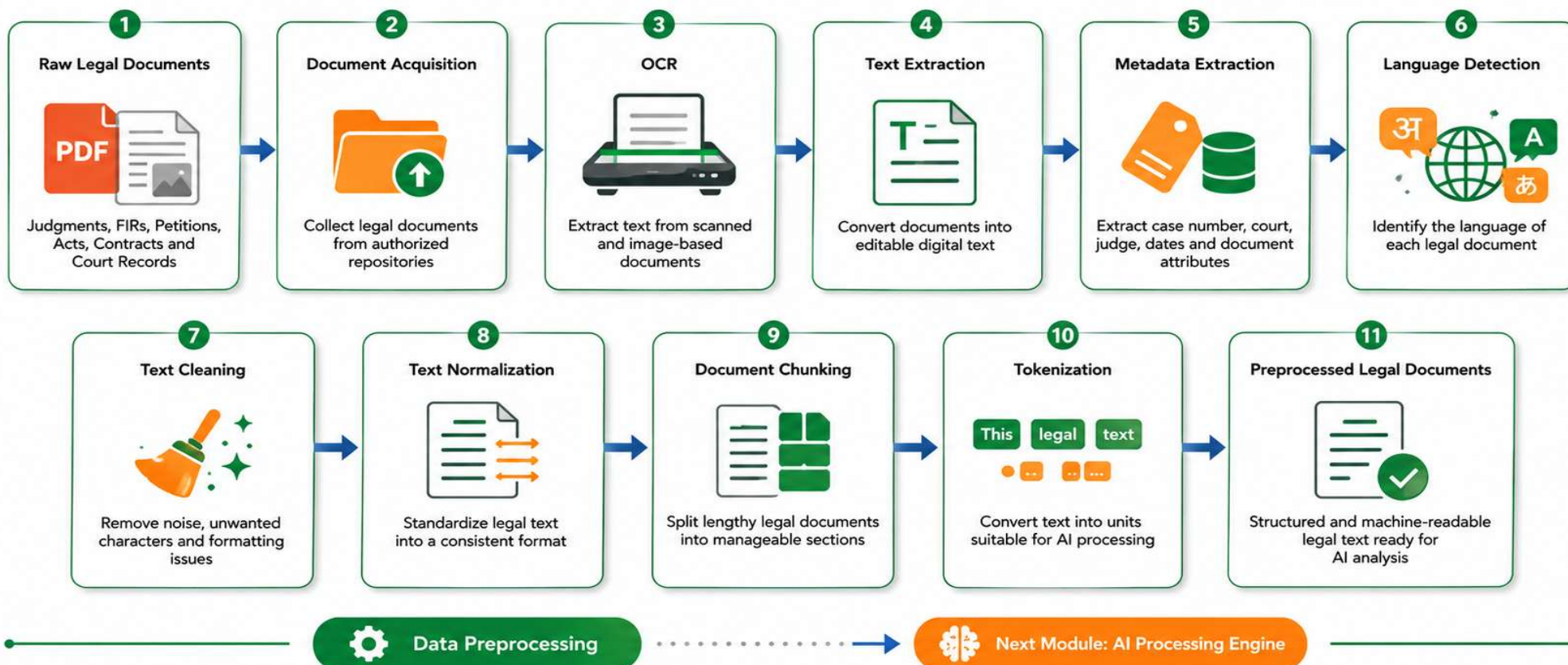
Module 1: Legal Documents (Input Layer)

Various legal document sources that serve as input to the AI-powered system



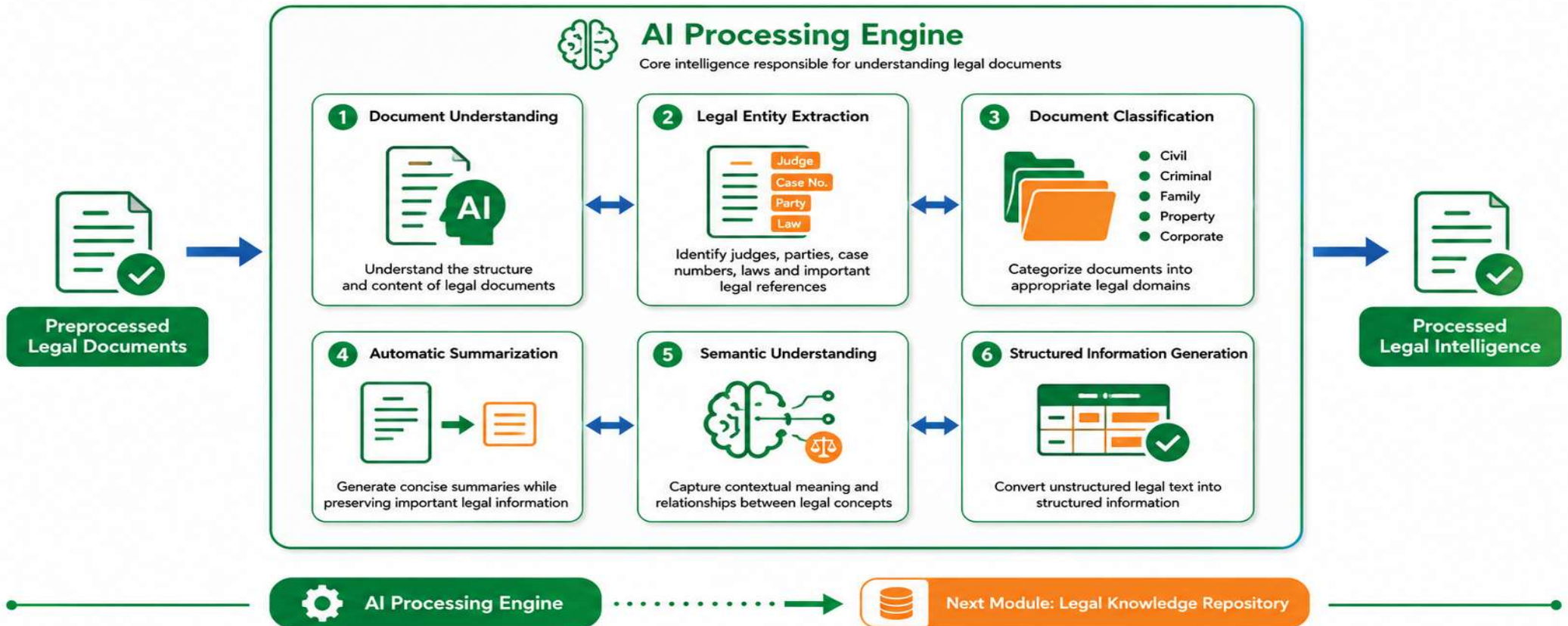
Module 2: Data Preprocessing

Preparing legal documents for AI-based analysis



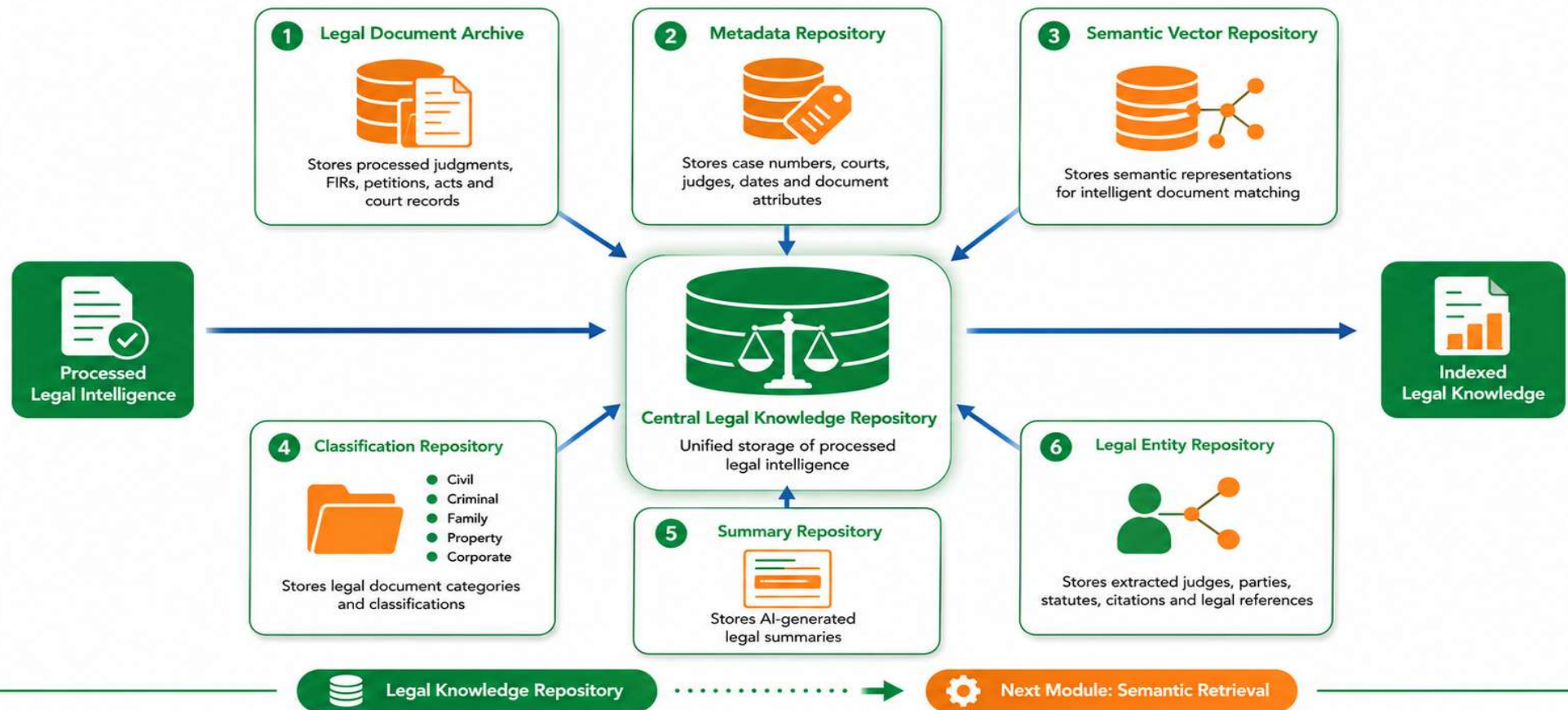
Module 3: AI Processing Engine

Intelligent analysis of legal documents using Artificial Intelligence and Natural Language Processing



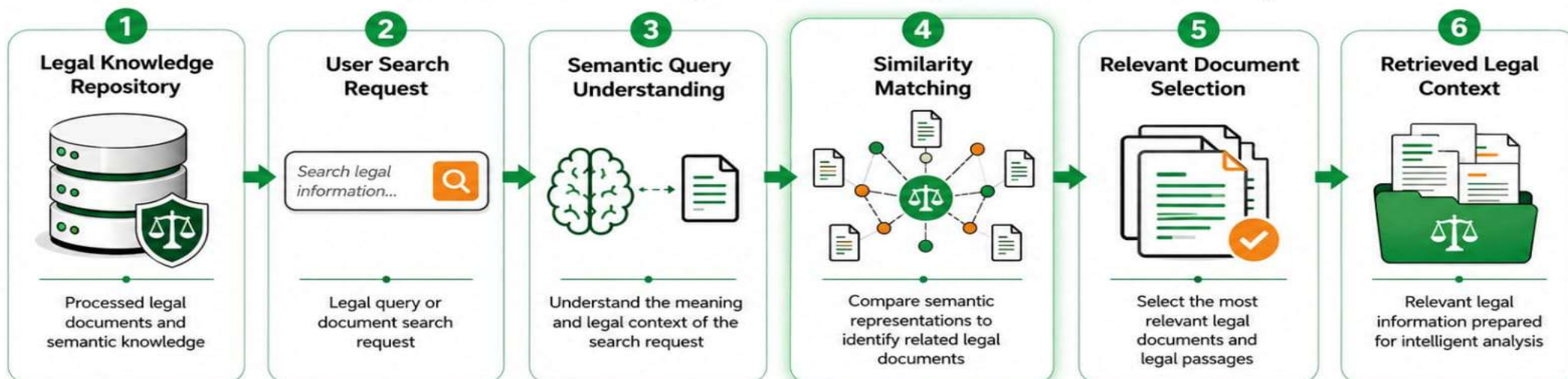
Module 4: Legal Knowledge Repository

Organizing processed legal information for efficient access and retrieval



Module 5: Semantic Retrieval

Finding the most relevant legal information using semantic understanding



Why Semantic Retrieval?



Keyword Search

- ✗ Exact word matching
- ✗ Limited context
- ✗ Misses related legal concepts

VS



Semantic Retrieval

- ✓ Meaning-based matching
- ✓ Context-aware retrieval
- ✓ Finds conceptually related legal documents



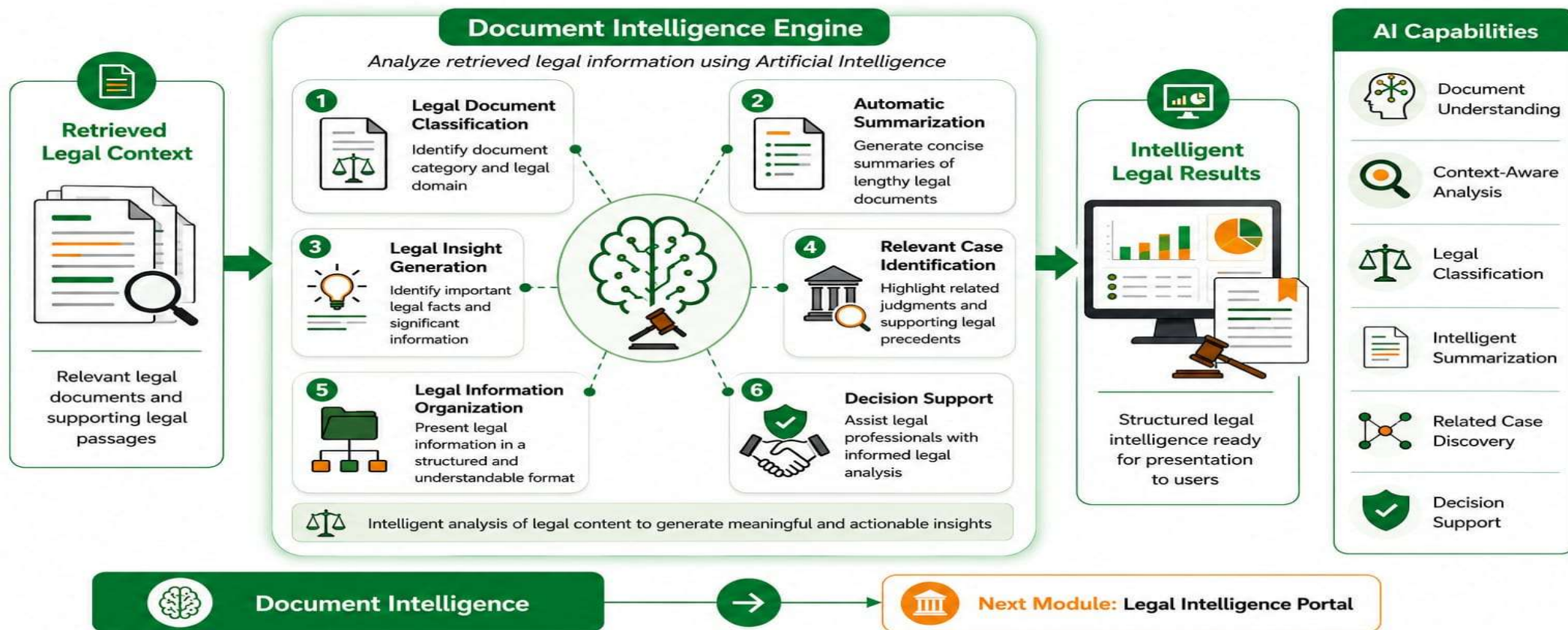
Semantic Retrieval



Next Module: Document Intelligence

Module 6: Document Intelligence

Transforming retrieved legal information into actionable legal insights





Module 7: Legal Intelligence Portal

Interactive platform for accessing AI-powered legal insights

Legal Intelligence Portal



Unified platform for
exploring AI-generated
legal intelligence

1 Smart Legal Search

Search...

Search legal documents
using natural language
queries

2 Case Summaries

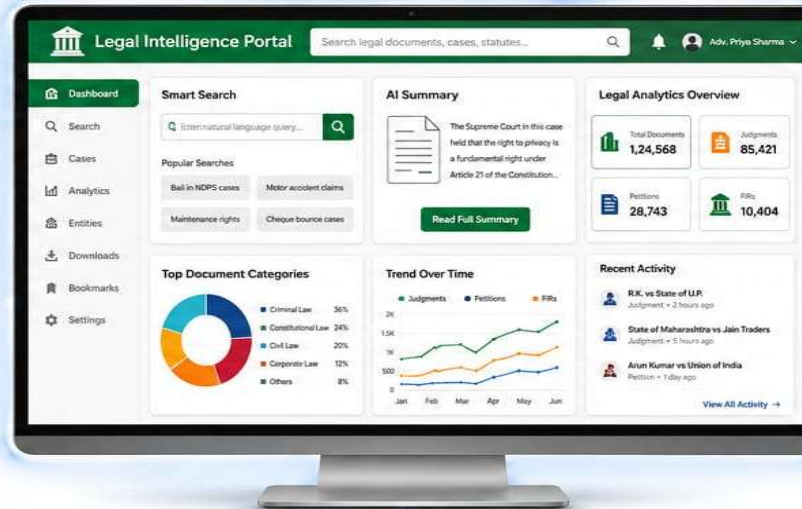
Quick access to
AI-generated case
summaries

3 Document Viewer

View complete
judgments, petitions,
FIRs and legal
documents

4 Related Case Explorer

Discover similar
judgments and
legal precedents



5 Legal Analytics Dashboard



Visualize legal
trends, document
categories and
insights

6 Entity Explorer

Browse extracted
legal entities and
relationships

7 Download & Export

Export reports,
summaries and
legal documents

8 Secure Access

Role-based
authentication
and secure access
to legal information

PORTAL HIGHLIGHTS

- ✓ Unified Legal Dashboard
- ✓ Intelligent Search
- ✓ AI-Generated Summaries
- ✓ Related Case Discovery
- ✓ Interactive Analytics
- ✓ Secure Document Access

Legal Information
for Users
(To Next Module)



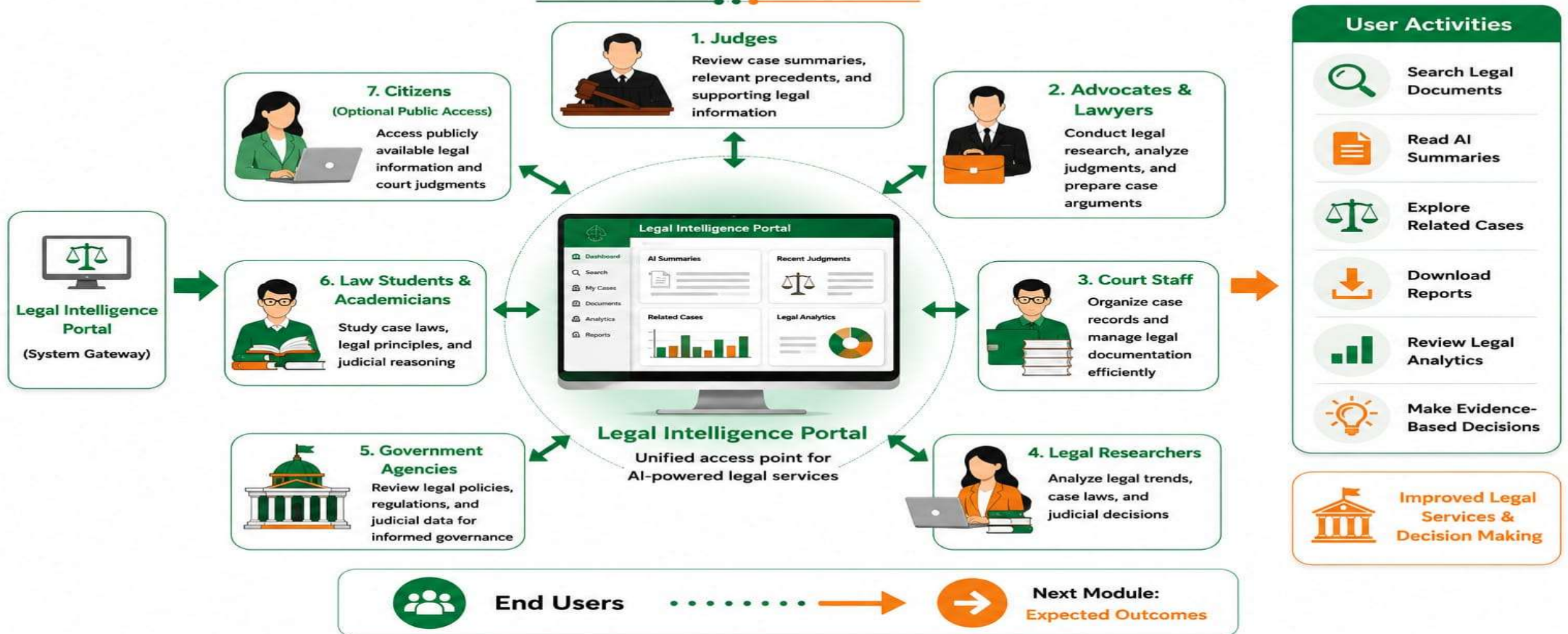
Legal Intelligence
Portal



Next Module:
End Users

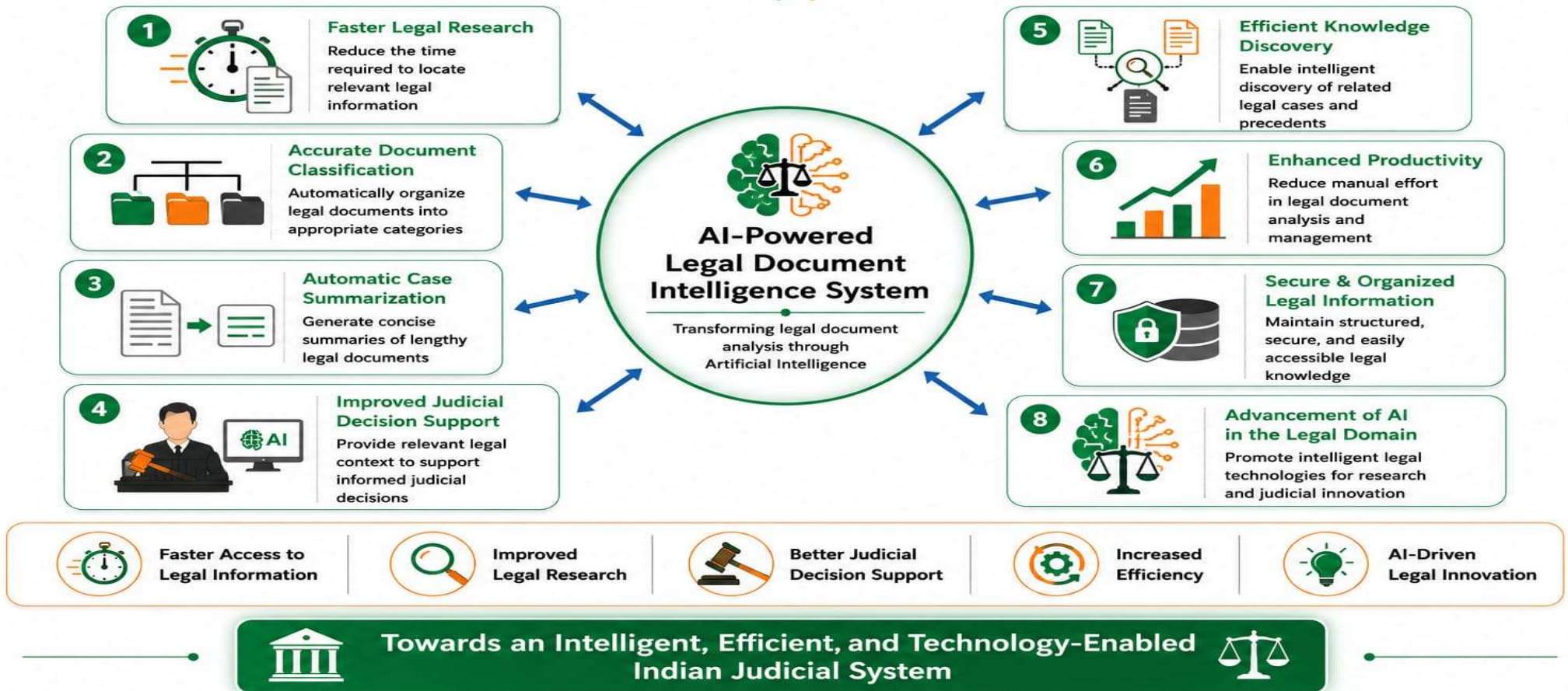
Module 8: End Users

Stakeholders benefiting from AI-powered legal intelligence



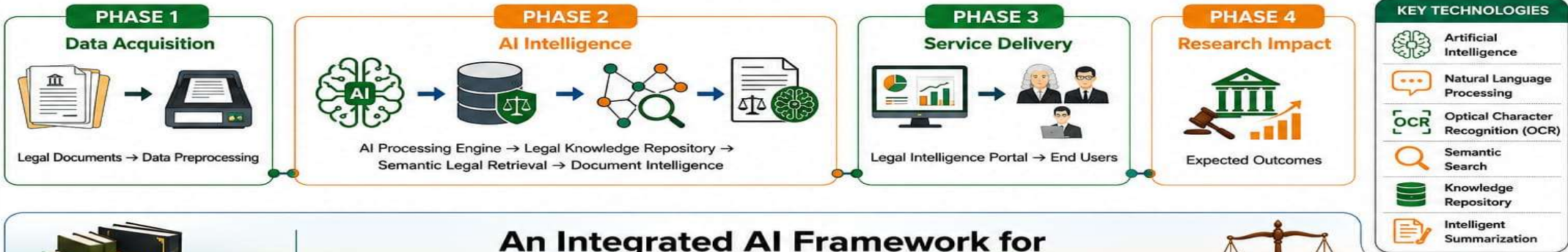
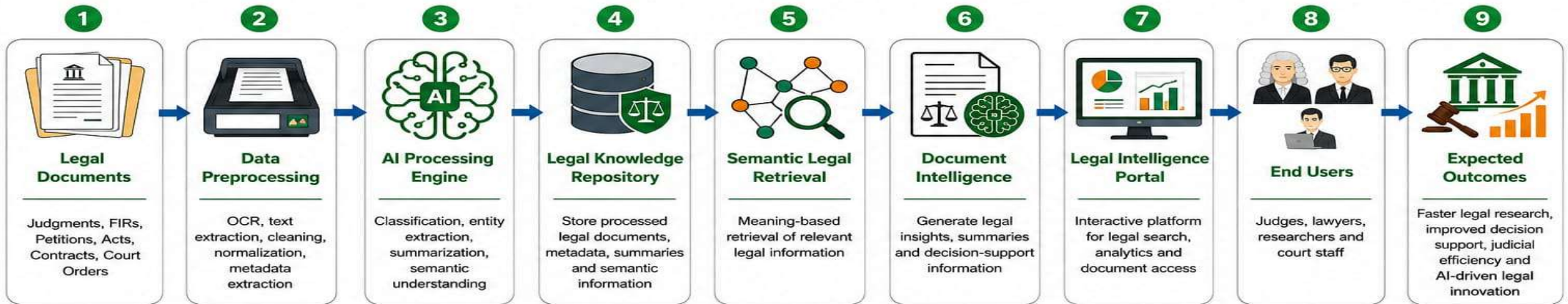
Module 9: Expected Outcomes

Expected impact of the proposed AI-powered legal document intelligence system



End-to-End System Overview

Workflow of the Proposed AI-Powered Legal Document Intelligence System for Indian Courts

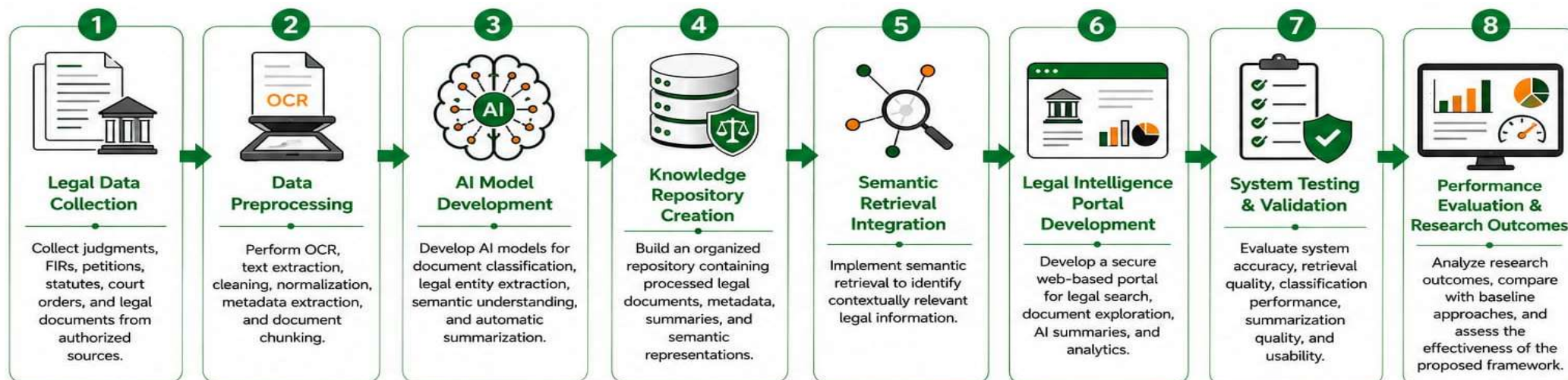


An Integrated AI Framework for
Intelligent Legal Document Analysis, Retrieval,
and Decision Support in Indian Courts.



Research Methodology

Methodology adopted for developing the AI-Powered Legal Document Intelligence System



Research Components

-  Legal NLP
-  OCR
-  Artificial Intelligence
-  Semantic Search
-  Knowledge Repository
-  Intelligent Summarization
-  Web Application
-  Performance Evaluation

Evaluation Parameters



Classification Accuracy



Summarization Quality



Retrieval Precision



Retrieval Recall



Response Time



User Satisfaction

Key Outcomes



Intelligent legal document analysis



Improved legal research efficiency



Enhanced decision support for legal professionals

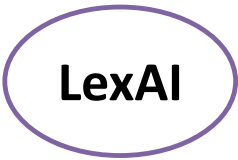


Scalable and adaptive AI framework for Indian Courts



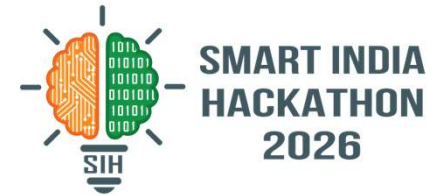
Develop, validate, and evaluate an integrated AI framework for intelligent legal document analysis and decision support in Indian Courts.



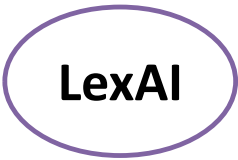


Research, References & Competitive Gap

A Comparative Study of Legal NLP & AI Methodologies



Research Paper	Core Methodology	Identified Research Gaps
Reducing Hallucinations in Legal AI: A Retrieval-Augmented Generation-Based Model for Accurate Legal Guidance S. Gupta, V. Agrawal, Y. Negi, S. Karunanithi, and A. Balakrishnan, IEEE Access, vol. 14, pp. 44451–44463, Mar. 2026, doi: 10.1109/ACCESS.2026.3675624	• PDF to Markdown segmentation into vector embeddings. • Vector similarity search contexts supplied to Llama-3.2-3B & Qwen2.5-3B. • Evaluated via ROUGE, BLEU, MPNet, BERTScore F1, and MRR.	• Focuses strictly on QA, omitting classification & summarization. • Tested on small dataset; lacks large-scale repository validation. • Multilingual & multimodal inputs remain unexplored.
Efficient and Fair Legal Judgment Classification via Personalized GraphRank Summarization and Class-Balanced Learning K. S. Rosamma, N. Patil, and C. D. Jaidhar, IEEE Access, vol. 14, pp. 104077–104089, 2026, doi: 10.1109/ACCESS.2026.3708100	• Personalized GraphRank (PGR) extractive summarization (TF-IDF + PageRank + MMR). • Multi-label classification using DeBERTa-v3-base. • Class-weighted Binary Cross-Entropy on LexGLUE (ECTHR).	• Concentrates on European law (ECTHR); lacks Indian legal context. • No Multi-lingual support for regional languages. • Missing an end-to-end unified legal management platform.
Legal Natural Language Processing from 2015-2022: A Comprehensive Systematic Mapping Study of Advances and Applications E. Quevedo, T. Cerny, A. Rodriguez, P. Rivas, J. Yero, K. Sookasatra, A. Zhakubayev, and D. Taibi, IEEE Access, vol. 11, Art. no. 3333946, 2023, doi: 10.1109/ACCESS.2023.3333946	• Systematic Mapping Study (SMS) analyzing 75 primary papers out of 536. • Maps techniques across QA, IR, Summarization, and IE. • Reviews SVM, Naive Bayes, BERT, Legal-BERT, Lawformer, T5, and GPT.	• Dominately English & Chinese centric. • Acute shortage of public datasets for low-resource Indic languages. • Absence of unified platforms integrating IR, classification, and summarization.



Research, References & Competitive Gap

A Comparative Study of Legal NLP & AI Methodologies



Research Paper	Core Methodology	Identified Research Gaps
Exploring LLMs Applications in Law: A Literature Review on Current Legal NLP Approaches M. Siino, M. Falco, D. Croce, and P. Rosso, IEEE Access, vol. 13, pp. 18253–18276, Jan. 2025, doi: 10.1109/ACCESS.2025.3533217	- PRISMA-guided Systematic Literature Review (61 selected papers). - Assesses LLMs, domain-adapted models, cross-lingual mechanisms. - Analyzes classification, legal reasoning, and QA pipelines.	- Low-resource multilingual processing remains severely underexplored. - Context window limits for ultra-long legal documents. - Unresolved explainability, ethical, and deployment barriers.
CHRExpert: An AI-Driven Court of Human Rights Expert Assistant for Legal Practitioners Utilizing Transformer Models Y. Al-Shareef, IEEE Access, vol. 13, pp. 41097–41110, 2025, doi: 10.1109/ACCESS.2025.3547763	~11,000 ECHR judgments cleaned, BPE tokenized. - Fine-tuned GPT-J transformer with multi-head attention & Adam optimizer. - Tasks: Issue spotting, precedent comparison, outcome prediction.	- ECHR specific; non-transferable to Indian legal systems without domain adaptation. - Lacks integrated semantic search & summarization. - Vulnerable to historical precedent bias & judicial discretion nuances.
SmartLaw Advisor: AI-Powered Legal Consultation and Contract Analysis for Saudi Arabia F. Almutairy, S. Almutairi, M. Abdullah, and S. Salman, IEEE Access, vol. 14, pp. 32638–32653, Mar. 2026, doi: 10.1109/ACCESS.2026.3524673	- Semantic chunking stored in ChromaDB vector store. - Cross-encoder re-ranking + Google Gemini 1.5 Flash RAG pipeline. - OCR integration for Arabic/English scanned legal contracts.	- Built specifically for Saudi jurisdiction; unusable for Indian law directly. - Scarce domain benchmarks for regional retrieval performance. - Focused on consultation/contracts rather than full workflow integration.